Traffic Analysis

In this project, we will:

- Load the dataset
- Understand the data
- Clean missing values
- Create beginner-friendly charts
- Generate simple insights

Step 1 — Import Libraries

```
import pandas as pd
import matplotlib.pyplot as plt

%matplotlib inline
```

Step 2 — Load Dataset

```
df = pd.read_csv("traffic.csv")

# Show first 5 rows
df.head()
```

	event	date	country	city	artist	album	track	isrc	linkid
0	click	2021-08-21	Saudi Arabia	Jeddah	Tesher	Jalebi Baby	Jalebi Baby	QZNWQ2070741	2d896d31-97b6-4869-967b-1c5fb9cd4bb8
1	click	2021-08-21	Saudi Arabia	Jeddah	Tesher	Jalebi Baby	Jalebi Baby	QZNWQ2070741	2d896d31-97b6-4869-967b-1c5fb9cd4bb8
2	click	2021-08-21	India	Ludhiana	Reyanna Maria	So Pretty	So Pretty	USUM72100871	23199824-9cf5-4b98-942a-34965c3b0cc2
3	click	2021-08-21	France	Unknown	Simone & Simaria, Sebastian Yatra	No Llores Más	No Llores Más	BRUM72003904	35573248-4e49-47c7-af80-08a960fa74cd
4	click	2021-08-21	Maldives	Malé	Tesher	Jalebi Baby	Jalebi Baby	QZNWQ2070741	2d896d31-97b6-4869-967b-

Step 3 — Check Dataset Information

```
# Dataset information
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 226278 entries, 0 to 226277
Data columns (total 9 columns):
 #   Column   Non-Null Count   Dtype
---  ------   --------------   -----
 0   event    226278 non-null  object
 1   date     226278 non-null  object
 2   country  226267 non-null  object
 3   city     226267 non-null  object
 4   artist   226241 non-null  object
 5   album    226273 non-null  object
```

```
 6   track    226273 non-null  object
 7   isrc     219157 non-null  object
 8   linkid   226278 non-null  object
dtypes: object(9)
memory usage: 15.5+ MB
```

Step 4 — View Column Names

```
# Show all column names
print(df.columns)
```

```
Index(['event', 'date', 'country', 'city', 'artist', 'album', 'track', 'isrc',
       'linkid'],
      dtype='object')
```

Step 5 — Check Missing Values

```
# Missing values
df.isnull().sum()
```

	0
event	0
date	0
country	11
city	11
artist	37
album	5
track	5
isrc	7121
linkid	0

dtype: int64

Step 6 — Remove Duplicate Rows

```
# Remove duplicates
df = df.drop_duplicates()

print("Duplicates removed successfully")
```

```
Duplicates removed successfully
```

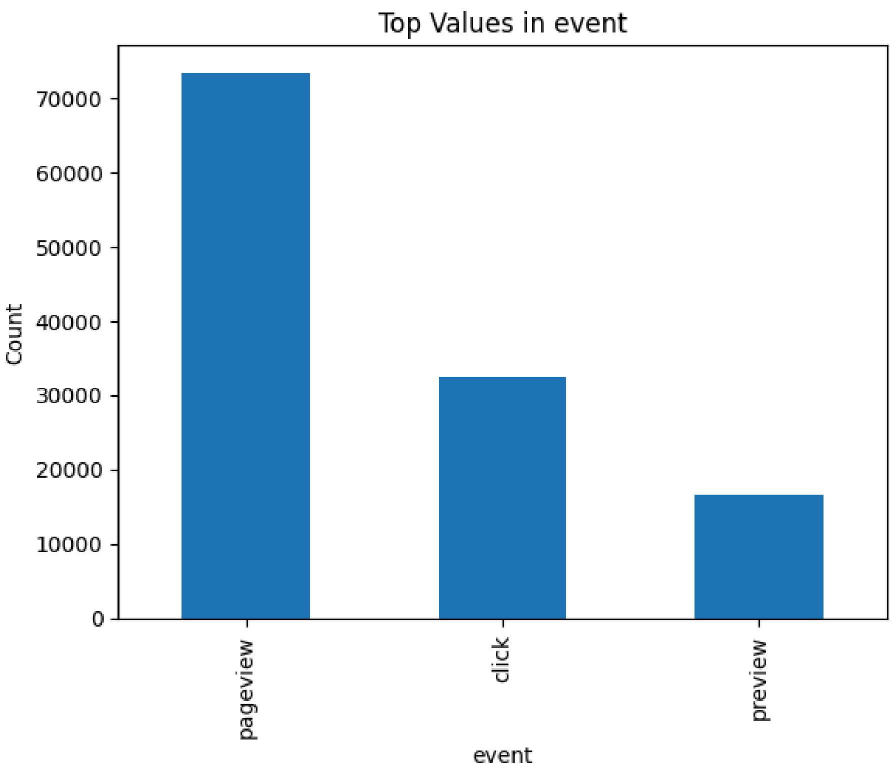
Step 9 — Top Categories

```
# Top categories
top_values = df['event'].value_counts().head(10)

top_values.plot(kind='bar')

plt.title('Top Values in event')
plt.xlabel('event')
plt.ylabel('Count')

plt.show()
```



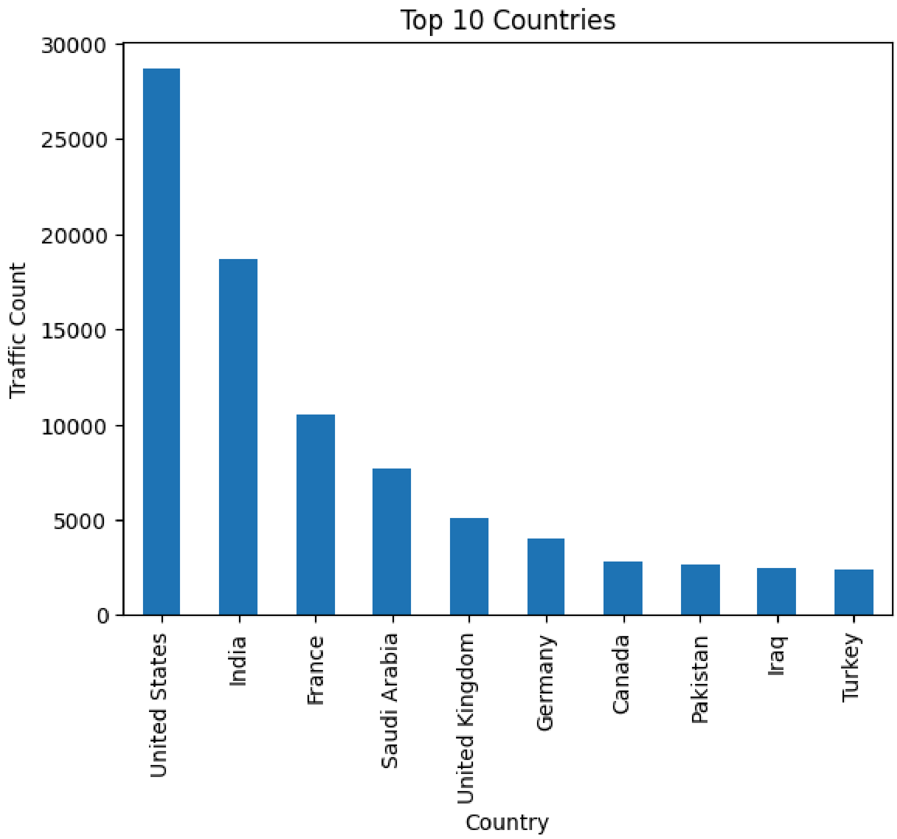
Step 9.1 Top 10 Countries by Traffic

```
# Top 10 countries
top_countries = df['country'].value_counts().head(10)

# Bar chart
top_countries.plot(kind='bar')

plt.title("Top 10 Countries")
plt.xlabel("Country")
plt.ylabel("Traffic Count")

plt.show()
```



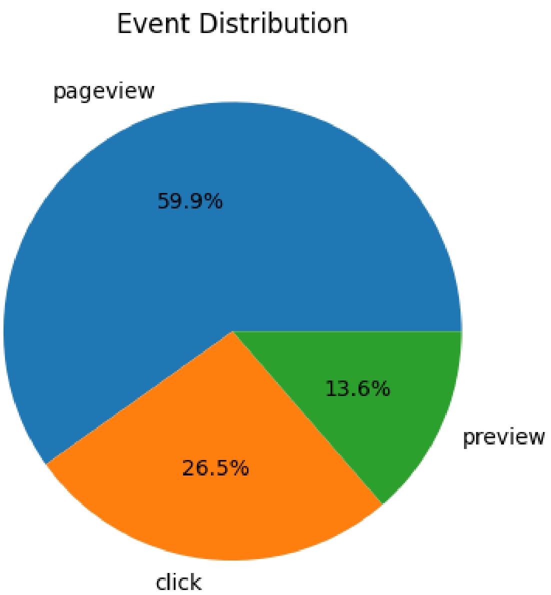
Step 9.2 -Event Type Distribution

```
# Count event types
event_counts = df['event'].value_counts()

# Pie chart
event_counts.plot(kind='pie', autopct='%1.1f%%')

plt.title("Event Distribution")
plt.ylabel("")

plt.show()
```



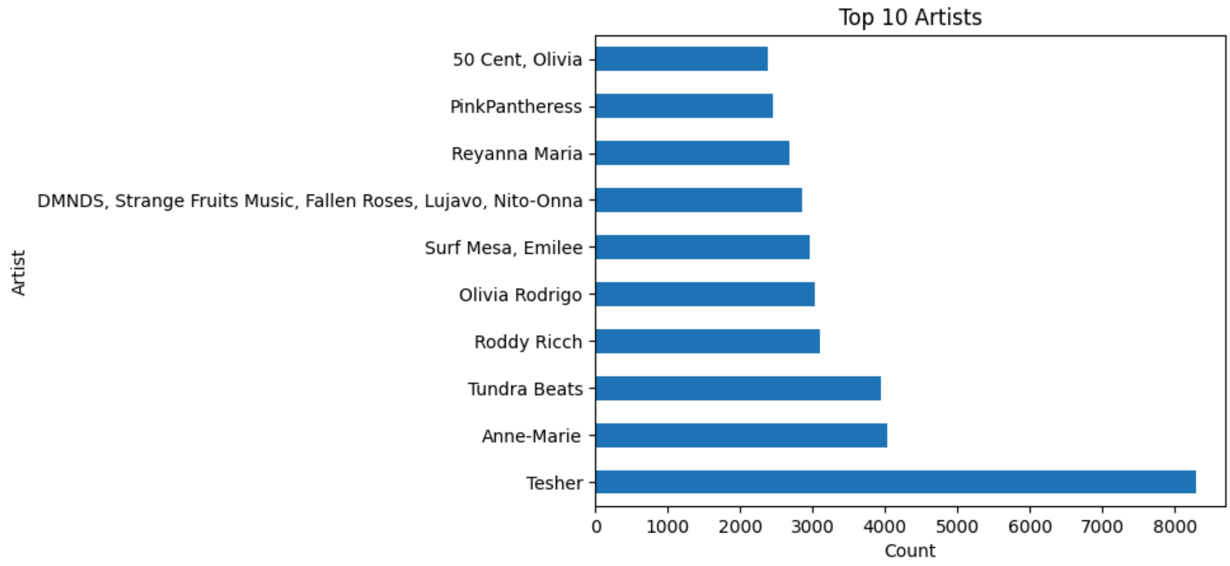
Step 9.3-Top 10 Artists

```
# Top artists
top_artists = df['artist'].value_counts().head(10)

# Horizontal bar chart
top_artists.plot(kind='barh')

plt.title("Top 10 Artists")
plt.xlabel("Count")
plt.ylabel("Artist")

plt.show()
```



Step 9.4- Top 10 Most Played Songs

```
# Top 10 Most Played Songs
top_songs = df['track'].value_counts().head(10)

# Horizontal bar chart
top_songs.plot(kind='barh')

plt.title("Top 10 Most Played Songs")
plt.xlabel("Number of Plays")
plt.ylabel("Song Name")

plt.show()
```